

Chapter XX

Carbon-Conscious Urban Heat Action through Remote Sensing-Based Assessment of Land-Specific Carbon Dynamics

Md. Naimur Rahman¹, Mst Ilme Faridatul¹, Md. Habibullah Masbah¹ *, Md. Omar Faruq¹, Md. Ubaidullah¹

¹ Department of Urban & Regional Planning, Rajshahi University of Engineering & Technology (RUET), Rajshahi 6204, Bangladesh

Correspondence email: habibullah.ruet.urp@gmail.com

Abstract

Rapid urbanisation in secondary cities of the Global South is converting carbon-absorbing land covers into emission-intensive built surfaces while intensifying surface heat, yet the spatial coupling of these two pressures is rarely resolved at a scale that urban planners can act upon. This chapter presents an urban heat action story for Rajshahi City Corporation (RCC), Bangladesh, in which land-specific carbon emission (LSCE) accounting is integrated with land surface temperature (LST) to identify where carbon stress and heat exposure coincide between 2000 and 2025, and in which these findings are converted into a practical action framework. Land-use/land-cover maps derived from Landsat imagery at five-year intervals were converted to LSCE with class-specific coefficients, and pre-monsoon (March–May) LST, NDVI and NDBI composites were generated from Landsat Collection 2 Level-2 data in Google Earth Engine. All layers were aggregated to a 240 m analytical grid (861 cells), Getis-Ord Gi* hotspots were mapped for LSCE and LST, and the two results were combined into carbon–heat stress zones. Built-up land expanded from 15.08 to 26.83 km² (+77.9%), while vegetation contracted from 23.42 to 13.78 km² (–41.2%). As a result, the city shifted from a net carbon sink (-1.64×10^3 t CO₂ yr⁻¹ in 2000) to a net source ($+3.71 \times 10^3$ t CO₂ yr⁻¹ in 2025), crossing the neutrality threshold between 2010 and 2015, and the share of net-source grid cells rose from 35.2% to 67.0%. LSCE was positively associated with LST ($r = 0.18–0.58$), and carbon-source cells were 0.40–1.42 °C hotter than sink cells, with the gap widening after 2010. Since 2015, more than half of the heat-hotspot area has coincided with carbon hotspots, forming dual-stress zones of 2.2–4.3 km² in the central and riverside core, whereas cool-sink zones persisted mainly on the eastern fringe. NDBI was the strongest correlate of LST ($r = 0.65–0.86$), and exposed char land along the Padma emerged as a carbon sink that is nevertheless a heat source. Building on these results, the chapter proposes a four-step carbon-conscious heat action cycle (monitor, zone, act and review) that assigns each stress zone a portfolio of measures under three pillars: policy and governance, assessment and awareness, and tactical solutions. Priority actions include protecting eastern cool sinks as carbon-sending areas, greening and cooling dual-stress neighbourhoods, capping impervious expansion in carbon hotspots, and linking annual satellite

monitoring to ward-level heat warnings. A set of measurable indicators, with 2025 baselines, is provided to track progress. The framework relies on freely available data and can be transferred to other rapidly urbanising secondary cities.

Keywords: Land-specific carbon emission; Land surface temperature; Getis-Ord Gi*; Carbon-heat stress zones; Urban heat island; Remote sensing; Rajshahi; Heat action planning

1. Introduction

Cities bring people together along with the necessary infrastructure and energy consumption, and it is urban areas that account for the majority of global greenhouse gas emissions associated with energy use (Lwasa et al., 2022). The physical expansion that goes with urban growth also provides a second, less obvious route: when vegetation, cropland and wetlands are converted into impervious surfaces, land that had previously been absorbing carbon is lost and is replaced by land that supports activities which are intensive in terms of emissions (Cui et al., 2018; Zhou et al., 2021). This type of land use change also affects the surface energy balance by reducing evapotranspiration and causing more heat to be stored in built-up materials, so that changes in land cover have the effect of causing both a carbon imbalance and surface warming (Srivastava and Bharat, 2025). For secondary cities in the Global South that are undergoing rapid urbanisation, where both of these processes are taking place at the same time and where planning capacity is limited, it is essential to understand where these pressures overlap if climate action is to address both mitigation and adaptation.

Land-use-based carbon accounting has become a practical method for estimating the carbon effects of urban expansion in areas where city-level energy data are not available. In this method, the area of each type of land use is multiplied by an empirical emission or sequestration factor, resulting in an estimate of spatially specific land-specific carbon emissions (LSCE) in which built-up land and cropland function as sources and vegetation, water, and bare land act as sinks (Fang et al., 2007; Fattah et al., 2021a). The technique has been used at the provincial and metropolitan levels in China (Gui et al., 2023; Ye and Ming, 2023; Fan et al., 2024) and is being applied more and more in Bangladesh, having been employed to assess the carbon impact of urban expansion in Khulna (Fattah et al., 2021a, 2021b) and in the Rajshahi metropolitan area (Rahman et al., 2024). Recent studies have demonstrated that aggregating the LSCE values to regular grids shows intra-urban clustering that administrative boundaries hide (Li et al., 2023; Li et al., 2024; Liu et al., 2024).

At the same time, a great deal of research has employed Landsat thermal data to show that surface urban heat islands in cities in Bangladesh have become more intense, linking the rise in temperature to the loss of vegetation and water bodies as well as to the expansion of impervious surfaces (Dewan et al., 2021; Kafy et al., 2021; Imran et al., 2022; Sresto et al., 2022). In Rajshahi, more recent analyses have found statistically significant LST hotspots in the central and

western areas of the city by means of the Getis-Ord G_i^* statistic (Rahman and Mark, 2025). A smaller number of studies have started to make the connection between carbon emissions and heat, for instance by relating land-use-based carbon emissions to surface temperature in Khulna (Fattah et al., 2021a) and in Semarang, Indonesia (Dewa and Buchori, 2023), or by combining carbon emission hotspots with LST to produce composite carbon-thermal maps (Srivastava and Bharat, 2025).

There are still three gaps. The first is that the majority of studies involving carbon and heat look at total figures for the whole city or for entire social classes and fail to identify, with statistical confidence, the exact locations where carbon stress and heat exposure are concentrated together. The second is that analyses of land surface temperature trends often compare images taken on a single date but under different weather conditions, thereby mixing up actual changes with day-to-day fluctuations. Third, even though it is widely expected that cities will transition from being net carbon sinks to net carbon sources, the timing and the spatial extent of this shift have seldom been recorded for a secondary city in South Asia. A separate study defined the areas that send and receive carbon across the 370 km² Rajshahi Development Authority (RDA) region and found that the region as a whole was still a weakening net sink in 2025 (Rahman et al., under review). Nevertheless, this regional average hides the situation in the dense urban core, where carbon and heat pressures are expected to be most severe.

This chapter extends the previous work to the urban core of Rajshahi City Corporation (RCC) and adds a thermal dimension. It has three objectives:

(i) to measure how land-specific carbon emission (LSCE) and pre-monsoon surface temperature changed across RCC between 2000 and 2025; (ii) to identify where carbon emission and surface heat cluster together, and how strongly the two are related; and (iii) to turn these findings into a zone-based heat action framework that shows what should be done, where, by whom, and how progress can be tracked.

2. Study Area

Rajshahi City Corporation is the main administrative and functional centre of Rajshahi, the principal city in north-western Bangladesh (see Figure 1-1). The area of the corporation that has been studied here amounts to 46.93 km² and is situated on the northern side of the Padma (Ganges) River, which lies along its southern boundary and marks the international border with India. The land is a flat alluvial plain and has a tropical monsoon climate, during which the months from March to May before the monsoon are the hottest and driest of the year. Rajshahi has for a long time been regarded as one of the greenest cities in Bangladesh, but the expansion of educational, medical and administrative institutions has led to an increase in population and has caused orchards, ponds and agricultural land to be converted into residential and commercial areas (Kafy et al., 2021; Rahman et al., 2024). The city is thus a typical example of a secondary city in which ecological land is being taken up by urban development

at the same time as heat stress is rising (Dewan et al., 2021; Rahman and Mark, 2025).

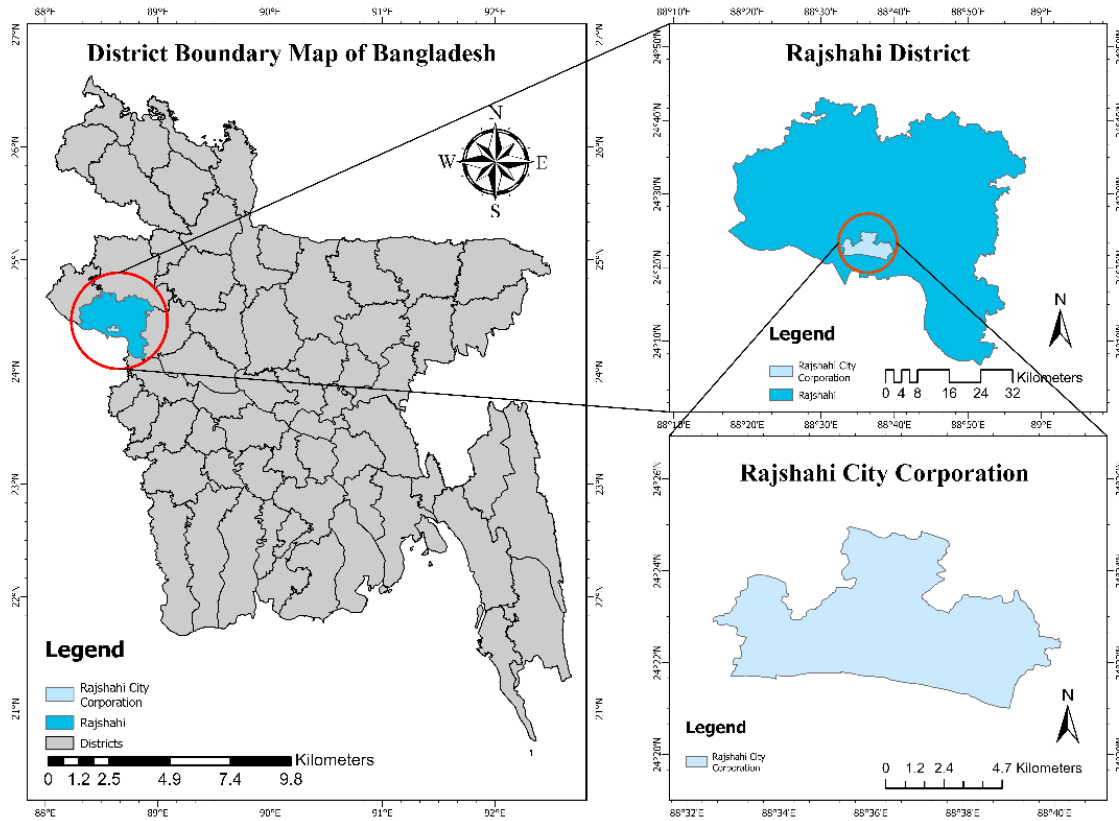


Figure 1-1. Location of the study area, Rajshahi City Corporation (RCC), Bangladesh.

3. Data and Methods

The analytical framework has three parallel components (Figure 1-2): a carbon pathway that converts land-use/land-cover (LULC) maps into LSCE, a thermal pathway that derives LST and spectral indices, and an integration pathway that aggregates both to a common grid, maps hotspots and classifies carbon–heat stress zones.

3.1 Satellite Data

For path/row 138/043 (as shown in Table 1-1) two sets of Landsat imagery were employed. The land use and land cover (LULC) was classified using the February scenes from the six benchmark years (2000, 2005, 2010, 2015, 2020 and 2025), a time when the skies were clear and there was little cloud cover. Since February does not correspond to the period of maximum heat stress, the thermal variables were calculated separately from all the cloud-screened Landsat Collection 2 Level-2 scenes obtained between 1 March and 31 May in each benchmark year (with a scene-level cloud cover of less than 30%). Landsat 5 TM was used for the years 2000 to 2010 and Landsat 8 OLI/TIRS and Landsat 9 OLI-2/TIRS-2 were used for the period from 2015 to 2025. The only Landsat 5 scene available for the year 2000 was cloud-covered over the city, so two Landsat 7 ETM+ scenes, which were acquired before the 2003 scan-line-corrector failure, were included for that year.

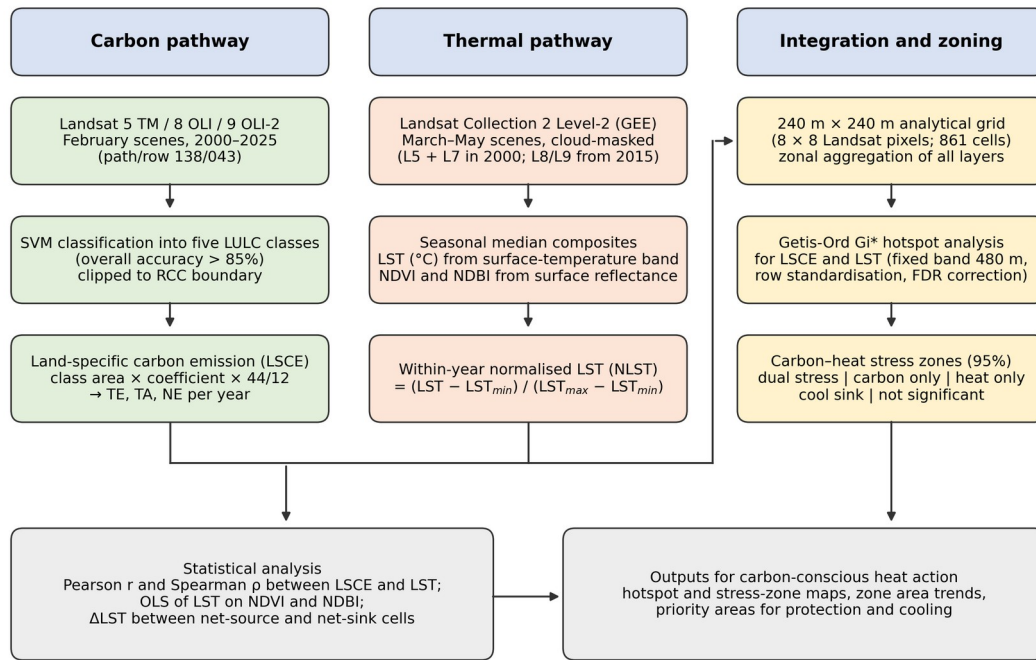


Figure 1-2. Methodological framework for integrating land-specific carbon emission and land surface temperature.

Table 1-1. Landsat imagery used in the study (path/row 138/043).

Year	LULC scene (sensor, date)	LST scenes, March–May (sensor, dates)	n
2000	L5 TM, 11 Feb 2000	L5 TM 15 Apr; L7 ETM+ 7 Apr, 9 May	3
2005	L5 TM, 24 Feb 2005	L5 TM 13 Apr, 15 May, 31 May	3
2010	L5 TM, 6 Feb 2010	L5 TM 11 Apr	1
2015	L8 OLI, 4 Feb 2015	L8 OLI 8 Mar, 24 Mar, 11 May, 27 May	4
2020	L8 OLI, 2 Feb 2020	L8 OLI 21 Mar, 6 Apr, 8 May	3
2025	L9 OLI-2, 7 Feb 2025	L8/L9 3 Mar, 11 Mar, 19 Mar, 27 Mar	4

Note: LULC scenes were obtained from the USGS EarthExplorer (cloud cover < 10%); LST scenes were accessed as Landsat Collection 2 Level-2 products in Google Earth Engine. n = number of LST scenes in the seasonal composite.

3.2 Land-Use/Land-Cover Classification

In February the scenes were divided into five categories—namely water body, vegetation, crop land, built-up area and bare/char land—by means of a support vector machine classifier. The training samples were obtained through visual interpretation of the imagery and by using high-resolution Google Earth images, and the chosen locations were then checked in the field with a GPS receiver. The classification accuracy was determined by means of an error matrix which was based on independent validation points; overall accuracy was greater than 85% for all six benchmark years. Classification had been carried out for the wider RDA area in the accompanying study (Rahman et al., under review) and the classified rasters were cut to match the RCC boundary in this chapter, the pixel grid being preserved by means of a snap raster so that all the years have the same 30 m lattice.

3.3 Land-Specific Carbon Emission

LSCE was estimated with the coefficient method (Fang et al., 2007; Fattah et al., 2021a; Rahman et al., 2024). For each class i , emission was calculated as:

$$E_i = S_i \times \delta_i \times (44/12) \quad (1)$$

where S_i is the area of class i , δ_i is the carbon emission coefficient of class i ($\text{kg C m}^{-2} \text{ yr}^{-1}$; Table 1-2) and $44/12$ converts elemental carbon to carbon dioxide. Positive coefficients denote sources and negative coefficients sinks. Summing the positive terms gives total emission (TE), summing the negative terms gives total absorption (TA), and their algebraic sum gives net emission (NE); a negative NE indicates that the landscape still functions as a net sink. Coefficients are expressed per square metre, which is the unit convention used in the coefficient literature (Li et al., 2024); with areas in km^2 , Equation (1) returns emissions in $10^3 \text{ t CO}_2 \text{ yr}^{-1}$.

Table 1-2. Land-use/land-cover classes and carbon emission coefficients.

LULC class	Description	Coefficient ($\text{kg C m}^{-2} \text{ yr}^{-1}$)	Role
Water body	Rivers, ponds, canals, marshes	-0.0459	Sink
Vegetation	Trees, orchards, urban green space	-0.0645	Sink
Crop land	Agricultural land and grassland	+0.0497	Source
Built-up area	Residential, commercial, industrial, roads	+0.0742	Source
Bare/char land	Open soil, playgrounds, char and sand bars	-0.0527	Sink

Source: adapted from Fang et al. (2007), Fattah et al. (2021a) and Rahman et al. (2024).

3.4 Land Surface Temperature and Spectral Indices

The surface-temperature values were derived from the surface-temperature band of the Landsat Collection 2 Level-2 products using a single-channel algorithm that includes atmospheric and emissivity corrections (Malakar et al., 2018). The digital numbers were rescaled to kelvin by means of the formula $\times 0.00341802 + 149.0$ and then converted into degrees Celsius. Clouds, diluted clouds and cloud shadows were identified using the QA_PIXEL band, and the other pixels from each benchmark year were combined to form a per-pixel median composite, this approach reducing the impact of individual acquisition dates and any remaining contamination (Qiu et al., 2023). The entire processing was performed in Google Earth Engine (Gorelick et al., 2017). The Normalised Difference Vegetation Index (NDVI; Tucker, 1979) and the Normalised Difference Built-up Index (NDBI; Zha et al., 2003) were calculated from the surface reflectance bands of the composites (reflectance = $\text{DN} \times 0.0000275 - 0.2$).

Since the dates of acquisition available from March to May varied from year to year (as shown in Table 1-1), the absolute LST values cannot be strictly compared between years. For this reason, the inter-annual comparisons were made using a within-year normalised LST (NLST), which was calculated for each grid cell by the formula $(\text{LST} - \text{LST}_{\min})/(\text{LST}_{\max} - \text{LST}_{\min})$, the minimum and maximum values referring to all the cells in that particular year. Similarly, the hotspot analysis described below involves a within-year comparison and is unaffected by the seasonal differences between years.

3.5 Grid-Based Integration

A 30 m pixel can have only one of five coefficient values, which means that it is impossible for pixel-level LSCE to represent the mixed land-use characteristics

of urban neighbourhoods. In order to address this, the study area was divided into a regular grid consisting of cells that are 240 m by 240 m, corresponding to 8 by 8 Landsat pixels, and the grid was aligned with the LULC pixel lattice so that no pixel would be split across different cells. The cell size is within the range that has been used in grid-based analyses of urban carbon emissions (Li et al., 2024) and is close to the 350 m resolution at which urban carbon emissions have been found to exhibit the strongest spatial autocorrelation (Liu et al., 2024). Any edge cells that were smaller than 25 per cent of a full cell were excluded, resulting in a total of 861 cells. For each cell, the LSCE density (in $\text{t CO km}^{-2} \text{ yr}^{-1}$) was calculated as the area-weighted average of the coefficients of the land cover classes present, multiplied by 44/12 and then converted into tonnes per km^2 , so that fully built-up and fully vegetated cells have values of +272.1 and -236.5 $\text{t CO}_2 \text{ km}^{-2} \text{ yr}^{-1}$ respectively. The mean values of LST, NLST, NDVI and NDBI were attached to each cell using zonal statistics after the no-data pixels had been removed.

3.6 Hotspot Analysis and Carbon–Heat Stress Zoning

For both LSCE and LST in each year, spatial clusters of high and low values were separately determined using the Getis-Ord G_i^* statistic (Getis and Ord, 1992; Ord and Getis, 1995), this method comparing the local sum of the value in a cell with that of its neighbours to the global mean and providing a z-score and p-value for each cell. Neighbourhoods were defined by a fixed distance range of 480 m (which is equivalent to two cell widths) with row standardisation, and the p-values were corrected for multiple testing and for spatial dependence using the false discovery rate procedure (Benjamini and Hochberg, 1995). Cells were designated as hot or cold spots at the 90%, 95% and 99% confidence levels. On the basis of the 95% threshold, each cell was then allocated to one of five carbon–heat stress zones: dual stress (meaning that the cell was a hotspot for both LSCE and LST), carbon hotspot only, heat hotspot only, cool sink (indicating that the cell was a coldspot for both LSCE and LST) and not significant. This approach of identifying hotspots in a statistical way is in line with standard procedures in urban thermal hotspot analysis (Guerri et al., 2021; Rahman and Mark, 2025) and extends those practices to include the carbon dimension.

3.7 Statistical Analysis

For each year the cell-level relationships were analysed using Pearson (r) and Spearman (ρ) correlation coefficients between LSCE and LST, as well as by means of ordinary least squares regression of LST on NDVI and NDBI, which are well-known biophysical predictors of surface temperature (Tiwari and Kanchan, 2024). The thermal contrast between carbon-source and carbon-sink cells was defined as the difference between the mean LST of cells having positive and negative LSCE (ΔLST). Since adjacent cells are spatially autocorrelated, the conventional p-values tend to overstate the level of significance; the results are therefore given in terms of effect sizes (r , ρ , R^2 , ΔLST) rather than as the results of hypothesis tests.

4. Results

4.1 Land-Use/Land-Cover Change

Over the 25-year period the land-use structure of RCC underwent a fundamental change (see Table 1-3). In the year 2000 vegetation was the main type of cover (23.42 km² or 49.9%) and built-up land took up about one-third of the city (15.08 km² or 32.1%). By 2025, built-up land had reached 26.83 km² (57.2%), a rise of 77.9%, while vegetation had dropped to 13.78 km² (29.4%), a decrease of 41.2%. This decline was not steady: vegetation fell sharply to 13.34 km² in 2015, then recovered to 17.51 km² in 2020 before falling once again by 2025, and at the same time crop land decreased from 3.64 to 2.19 km². The area occupied by water bodies stayed between 2.6 and 3.9 km², and bare/char land varied between 1.1 and 3.7 km² as a result of the seasonal exposure of sand bars along the Padma.

Table 1-3. Area of land-use/land-cover classes in RCC, 2000–2025 (km², with percentage share).

Class	2000	2005	2010	2015	2020	2025
Water body	2.61 (5.6)	2.96 (6.3)	3.84 (8.2)	3.39 (7.2)	3.21 (6.8)	3.04 (6.5)
Vegetation	23.42 (49.9)	22.56 (48.1)	20.94 (44.6)	13.34 (28.4)	17.51 (37.3)	13.78 (29.4)
Crop land	3.64 (7.7)	4.18 (8.9)	2.33 (5.0)	3.61 (7.7)	1.15 (2.5)	2.19 (4.7)
Built-up area	15.08 (32.1)	15.87 (33.8)	18.22 (38.8)	23.33 (49.7)	21.37 (45.5)	26.83 (57.2)
Bare/char land	2.19 (4.7)	1.36 (2.9)	1.59 (3.4)	3.26 (6.9)	3.68 (7.8)	1.10 (2.3)
Total	46.93	46.93	46.93	46.93	46.93	46.93

4.2 Land-Specific Carbon Emission Dynamics

Between 2000 and 2025 the amount of built-up land increased and as a result total emission rose from $4.76 \times 10^3 \text{ t CO}_2 \text{ yr}^{-1}$ to $7.70 \times 10^3 \text{ t CO}_2 \text{ yr}^{-1}$ (an increase of +61.8%), almost entirely due to the built-up category, the contribution of which went up from 4.10 to $7.30 \times 10^3 \text{ t CO}_2 \text{ yr}^{-1}$ (see Table 1-4). During this same period total absorption declined from -6.40 to $-3.98 \times 10^3 \text{ t CO}_2 \text{ yr}^{-1}$ (a decrease of -37.8%) as a result of the loss of vegetation. As a consequence, the net emission changed from a sink to a source: in 2000 RCC was a net sink of $-1.64 \times 10^3 \text{ t CO}_2 \text{ yr}^{-1}$, the strength of the sink decreased to $-0.52 \times 10^3 \text{ t CO}_2 \text{ yr}^{-1}$ by 2010, and the city then became a net source between 2010 and 2015, reaching $+2.65 \times 10^3 \text{ t CO}_2 \text{ yr}^{-1}$ in 2015 and $+3.71 \times 10^3 \text{ t CO}_2 \text{ yr}^{-1}$ in 2025. The partial recovery of vegetation in 2020 reduced but did not reverse the source status ($+0.63 \times 10^3 \text{ t CO}_2 \text{ yr}^{-1}$).

Table 1-4. Land-specific carbon emission by class and net balance in RCC, 2000–2025 ($10^3 \text{ t CO}_2 \text{ yr}^{-1}$).

Component	2000	2005	2010	2015	2020	2025
Built-up area	4.10	4.32	4.96	6.35	5.81	7.30
Crop land	0.66	0.76	0.42	0.66	0.21	0.40
Vegetation	-5.54	-5.34	-4.96	-3.16	-4.14	-3.26
Water body	-0.44	-0.50	-0.65	-0.57	-0.54	-0.51
Bare/char land	-0.42	-0.26	-0.31	-0.63	-0.71	-0.21
Total emission (TE)	4.76	5.08	5.39	7.01	6.02	7.70
Total absorption (TA)	-6.40	-6.10	-5.90	-4.36	-5.39	-3.98
Net emission (NE)	-1.64	-1.02	-0.52	+2.65	+0.63	+3.71

Note: negative values denote carbon absorption; class values may not sum exactly to totals because of rounding.

At the grid level, the proportion of cells functioning as net sources rose from 35.2 per cent (303 cells) in the year 2000 to 64.0 per cent in 2015 and to 67.0 per cent (577 cells) in 2025, and the mean LSCE density increased from -38.3 to $+77.0 \text{ t CO}_2 \text{ km}^{-2} \text{ yr}^{-1}$. As shown in Figure 1-3, the high-emission cells were at first located in a band stretching west to east through the southern and central areas of the city. After 2010, however, this band expanded and moved northwards, and by 2025 emission-dominated cells had formed a nearly unbroken block covering the central part of the city, while the net-sink cells remained mainly on the eastern and north-eastern edges and in small areas along the south-western riverfront.

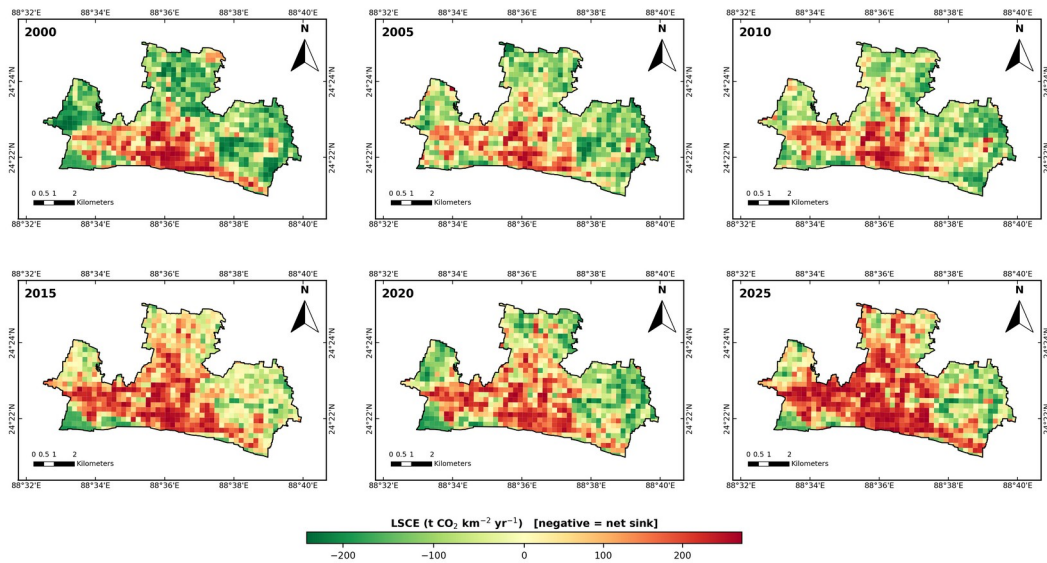


Figure 1-3. Land-specific carbon emission density per 240 m grid cell in RCC, 2000–2025.

4.3 Pre-Monsoon Thermal Patterns

The mean pre-monsoon land surface temperature (LST) over the RCC varied from 34.05°C in 2025 to 39.34°C in 2010, and at the cell level the temperatures ranged from about 27°C to 46.7°C . Since the annual composites were formed from different sets of March, April and May images (as shown in Table 1-1), these absolute temperature values represent both the seasonal timing and long-term changes; for instance, the 2025 composite included only images from March. The normalised surfaces (Figure 1-4) all display a similar internal structure: the highest temperatures were found in the central part of the city and along the southern riverbank areas that were exposed, while after 2005 the eastern edge became successively cooler in comparison with the rest of the city. The extreme values in the south-western corner are due to the river water and the exposed char land at the point where the city boundary meets the Padma.

4.4 Hotspots of Carbon Emission and Surface Heat

The Gi* analysis revealed persistent and statistically significant clustering of LSCE (Figure 1-5). A carbon hotspot occupied the central and west-central core in every year, while carbon coldspots were located on the eastern fringe and, in earlier years, in the western and northern peripheries. Between 2000 and 2025 the western and northern coldspots largely disappeared, indicating that the vegetated periphery that previously balanced the core had been converted to built-up use. LST hotspots were more dynamic (Figure 1-6). In 2000 and 2010

they were scattered across the northern and north-eastern peripheries, where exposed agricultural and bare land heated rapidly; from 2015 onwards they shifted towards the central core and the southern riverside, increasingly overlapping the carbon hotspot.

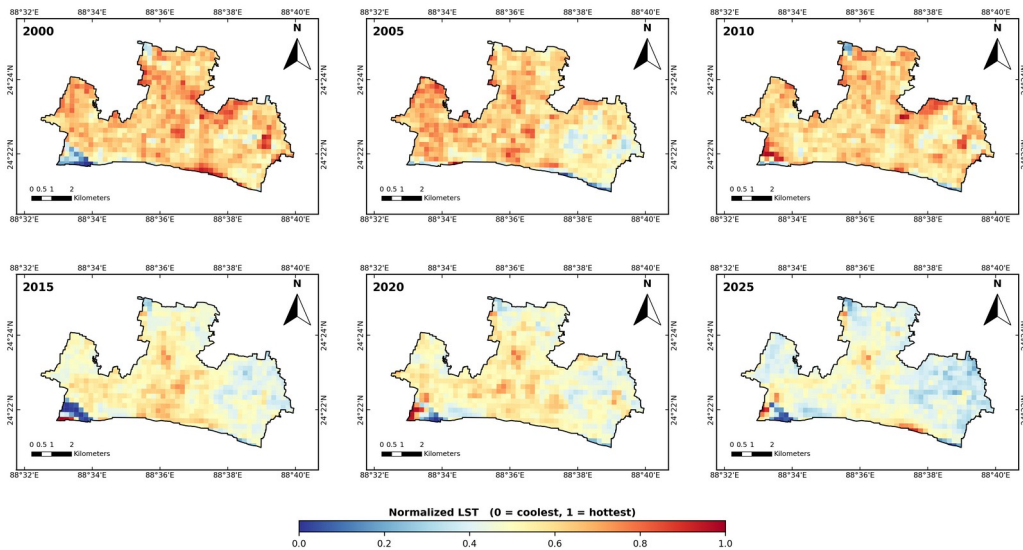


Figure 1-4. Within-year normalised pre-monsoon land surface temperature (NLST) in RCC, 2000–2025.

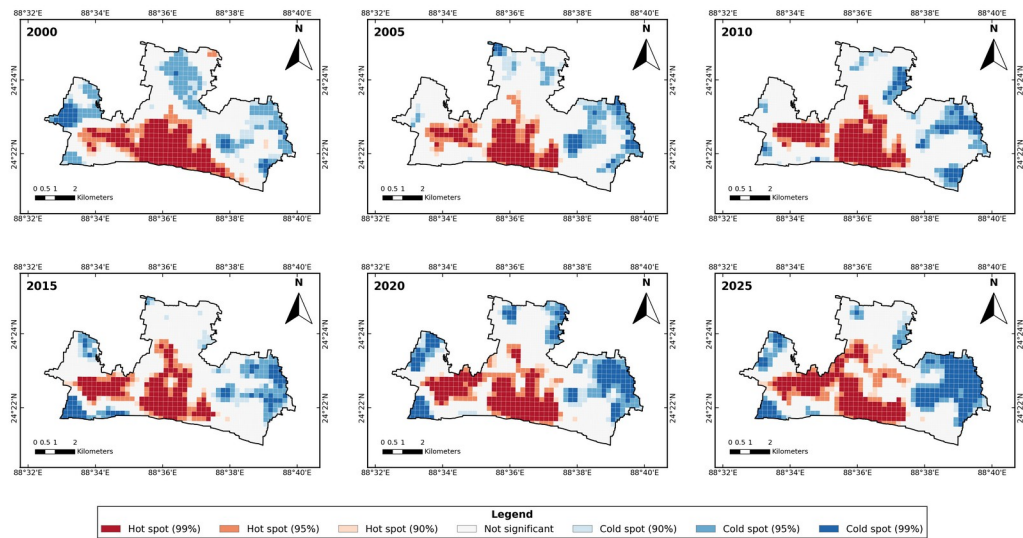


Figure 1-5. Hot and cold spots of land-specific carbon emission (LSCE) in RCC, 2000–2025 (Getis-Ord Gi*, FDR corrected).

4.5 Carbon–Heat Stress Zones

When the two hotspot surfaces are combined, the carbon–heat stress areas shown in Figure 1-7 and summarised in Table 1-5 and Figure 1-8 result. The carbon-only hotspots were the most extensive category in each year (ranging from 6.4 to 10.5 km²), thus confirming that carbon stress is more widespread than statistically significant heat stress. The dual-stress areas were small in 2000 (1.06 km²) and were not present in 2010, but increased to 4.32 km² in 2015 and stayed at 2.2 km² in both 2020 and 2025. More significantly, the proportion of the area of heat hotspots that also acted as a carbon hotspot increased from 27% in 2000 and 41% in 2005 to 89% in 2015, 57% in 2020 and 53% in 2025. That is to say, before 2010 heat hotspots were mainly a peripheral

phenomenon with no connection to the carbon core, while after 2010 most of the city's notable heat was concentrated in areas where carbon emissions were also concentrated. In 2025, the dual-stress areas formed two clusters: one in the north-central part of the city and another along the southern riverside. The cool-sink areas, in which low emissions occur together with low temperature, expanded from 0.66 km² in 2000 to 3.53 km² in 2025 and were located on the eastern fringe.

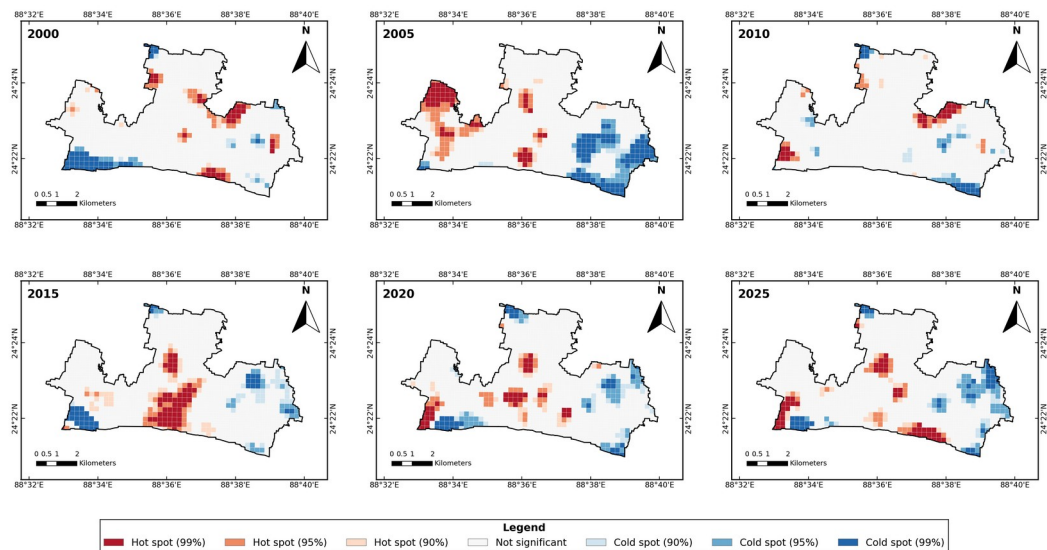


Figure 1-6. Hot and cold spots of land surface temperature (LST) in RCC, 2000–2025 (Getis-Ord Gi*, FDR corrected).

Table 1-5. Area of carbon–heat stress zones in RCC, 2000–2025 (km²).

Zone	2000	2005	2010	2015	2020	2025
Dual stress	1.06	2.40	0.00	4.32	2.25	2.18
Carbon hotspot only	10.43	7.17	10.51	6.43	9.40	9.59
Heat hotspot only	2.82	3.47	2.67	0.52	1.69	1.95
Cool sink	0.66	3.35	0.75	2.22	1.93	3.53
Not significant	31.53	30.11	32.58	33.00	31.24	29.24
Heat hotspots that are also carbon hotspots (%)	27	41	0	89	57	53

Note: areas refer to the 861 analytical cells (46.5 km²); edge cells smaller than 25% of a full cell were excluded.

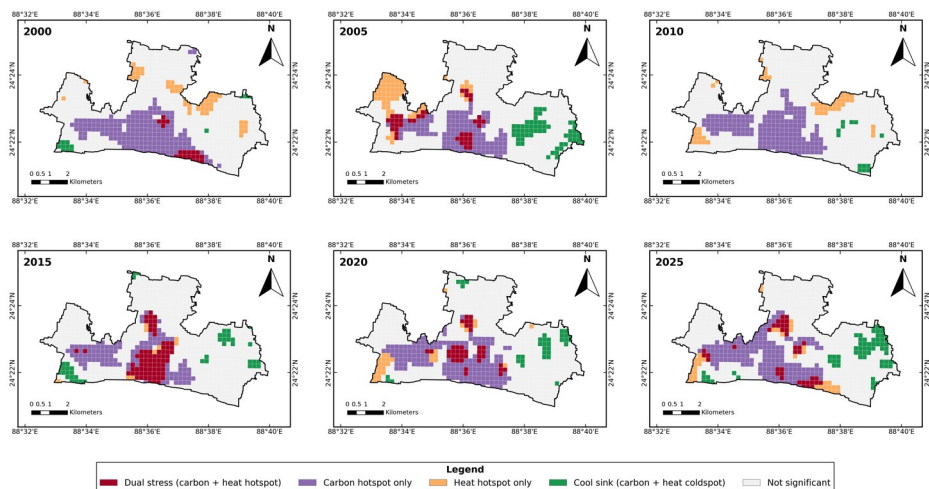


Figure 1-7. Carbon–heat stress zones in RCC, 2000–2025, derived from Gi* hotspots of LSCE and LST at 95% confidence.

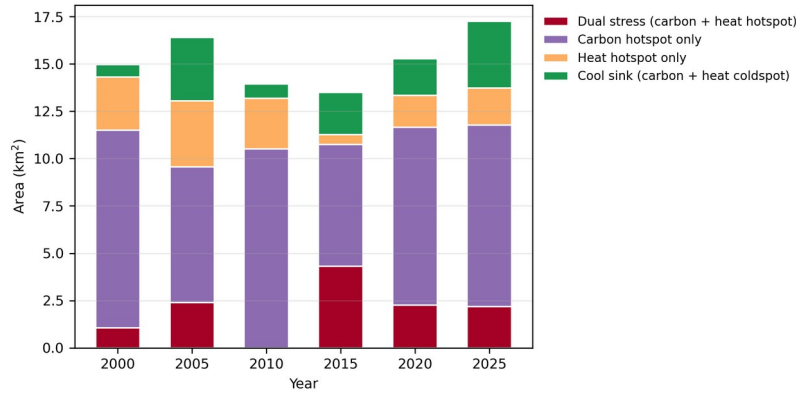


Figure 1-8. Temporal change in the area of carbon–heat stress zones in RCC, 2000–2025.

4.6 Relationship between Carbon Emission and Surface Temperature

LSCE and LST were positively correlated in every year (Table 1-6; Figure 1-9). The association was weak in 2000 ($r = 0.19$) and 2010 ($r = 0.18$) and moderate in 2005 ($r = 0.51$), 2015 ($r = 0.58$), 2020 ($r = 0.42$) and 2025 ($r = 0.51$); Spearman coefficients were consistently higher (ρ up to 0.68), indicating a monotonic but not strictly linear relationship. The thermal gap between net-source and net-sink cells increased from 0.55 °C in 2000 and 0.40 °C in 2010 to 1.42 °C in 2015, 1.35 °C in 2020 and 1.33 °C in 2025 (Figure 1-10b). Among the spectral indices, NDBI was by far the strongest correlate of LST ($r = 0.65$ – 0.86), and NDVI and NDBI together explained 55–83% of the variance in cell-level LST. NDVI alone showed weak and inconsistent associations ($r = -0.39$ to 0.03), and LSCE was moderately related to both NDVI ($r = -0.19$ to -0.49) and NDBI ($r = 0.29$ to 0.63).

Table 1-6. Cell-level relationships between LSCE, spectral indices and LST in RCC ($n = 861$).

Year	r (LSCE–LST)	ρ (LSCE–LST)	r (NDBI–LST)	r (NDVI–LST)	R^2 (LST ~ NDVI + NDBI)	Δ LST source – sink (°C)	Net-source cells (%)
2000	0.19	0.21	0.75	0.03	0.70	0.55	35.2
2005	0.51	0.53	0.65	-0.19	0.55	1.08	38.6
2010	0.18	0.22	0.85	-0.39	0.80	0.40	41.5
2015	0.58	0.66	0.86	0.03	0.83	1.42	64.0
2020	0.42	0.54	0.78	-0.24	0.69	1.35	49.6
2025	0.51	0.68	0.83	-0.35	0.76	1.33	67.0

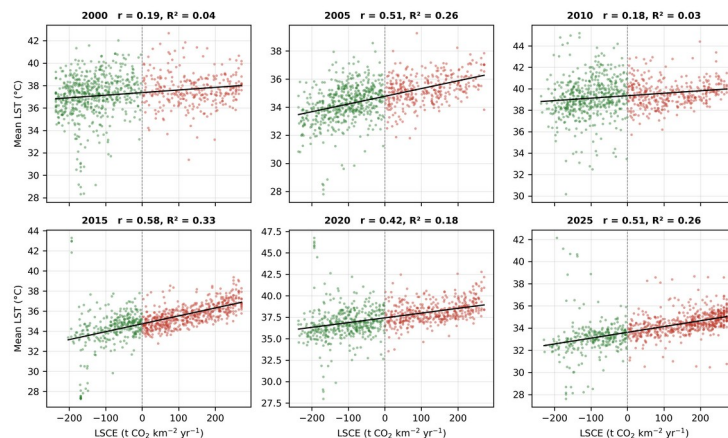


Figure 1-9. Relationship between land-specific carbon emission and mean LST of grid cells in RCC, 2000–2025 (red: net-source cells; green: net-sink cells).

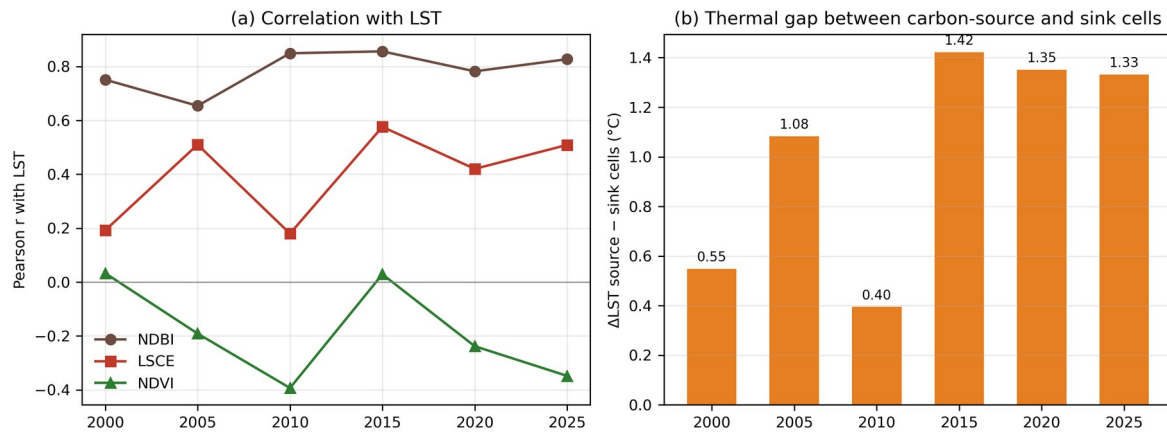


Figure 1-10. (a) Correlation of LSCE, NDBI and NDVI with LST; (b) difference in mean LST between net-source and net-sink cells, 2000–2025.

The scatter plots also reveal a distinct group of cells with strongly negative LSCE but high LST, visible in the upper-left of the 2010, 2015, 2020 and 2025 panels. These cells lie mainly along the Padma edge, where exposed sand and char land are classified as carbon sinks but heat intensely during the pre-monsoon season. They weaken the overall carbon–heat correlation and illustrate that carbon sinks and thermal refuges are not always the same land.

5. Discussion

5.1 From Net Sink to Net Source

The key result is that the urban area of Rajshahi changed from being a net carbon sink to a net carbon source between 2010 and 2015. This is in contrast to the broader RDA region, which in 2025 was still a weakening net sink (Rahman et al., under review), and shows how regional averages can mask the carbon status of dense urban areas. This shift was caused by the combined effect of increasing built-up emissions and a reduction in vegetation absorption, something that is in line with previous observations of vegetation loss and warming in Rajshahi (Kafy et al., 2021; Rahman and Mark, 2025) and with land-use carbon studies in other areas that point to constructed land as the main source and to forests and green land as the main sink (Gui et al., 2023; Li et al., 2024). The temporary recovery in 2020 proves that the balance is still affected by the amount of green cover, but the further decline by 2025 shows that the fundamental trend has not been reversed. For a city which has long been known for its green spaces, this is a clear indication that the existing vegetated land now has a disproportionately important role in its carbon balance.

5.2 Coupling and Decoupling of Carbon Stress and Heat

The fact that there is a positive and reinforcing relationship between LSCE and LST, along with the increasing thermal gap between source and sink cells, shows that carbon stress and heat exposure have become more closely linked as the city has grown denser. This is in agreement with research that has linked land-use-related emissions to surface temperature in Khulna and Semarang (Fattah et al., 2021a; Dewa and Buchori, 2023) and with the composite carbon–thermal mapping carried out in Indian cities (Srivastava and Bharat, 2025). The

spatial evidence is particularly clear in that, prior to 2010, the most significant heat was found on the agricultural fringe, but after 2010 more than half of the heat-hotspot area matched up with carbon hotspots. Since the peripheral farmland was developed over and the central area was consolidated, the thermal and carbon patterns of the city have come together.

The coupling is still only partial. This is evident from the moderate correlation coefficients, the fact that NDBI is more influential than LSCE as a predictor of LST, and the presence of areas that are hot but absorb carbon, all of which indicate that LSCE accounts for the carbon aspect of land cover but not for all of its thermal behaviour. Although bare sand has a negative carbon coefficient it still has a high surface temperature, and water has both a negative coefficient and a low temperature. The weak and inconsistent relationships between NDVI and LST also reflect the dry pre-monsoon conditions, during which crop land and sparse vegetation can be just as warm as built-up surfaces (Tiwari and Kanchan, 2024). Heat action with a focus on carbon therefore needs to make a distinction between land that acts as both a carbon sink and a thermal refuge, for example tree cover and water, and land that is a carbon sink but at the same time a heat source, such as exposed char.

5.3 Implications for Carbon-Conscious Urban Heat Action

The stress-zone maps translate these findings into spatial priorities. Dual-stress zones in the north-central core and along the southern riverside are the locations where interventions can deliver both mitigation and adaptation benefits: street-tree planting, pocket parks, restoration of ponds and high-albedo or vegetated roofs would simultaneously add sink capacity and reduce surface temperature. Evidence from European cities shows that urban tree cover can reduce land surface temperature substantially relative to built surfaces (Schwaab et al., 2021). Carbon-only hotspots, which cover the largest stressed area, call for measures aimed at the emission side, including limits on further impervious expansion, compact and mixed-use redevelopment and requirements to replace lost vegetation within development schemes. Cool-sink zones on the eastern fringe represent the city's remaining carbon-and-heat refuge and should be protected from conversion; in the language of the companion study, they are candidate carbon-sending areas whose development rights could be transferred to receiving areas in the consolidated core (Rahman et al., under review). Finally, because LSCE and LST hotspots can be updated from freely available Landsat data, the framework can support routine carbon and heat screening of development proposals by the city corporation and the development authority.

5.4 Limitations and Future Research

It is necessary to recognise a number of limitations. The emission coefficients are generalised figures based on data from the literature rather than being the result of local calibration, and the built-up coefficient does not make a distinction between residential, commercial and industrial usage; it follows that the absolute LSCE values must be regarded as indications of the carbon pressure resulting from land use rather than as a full urban emission inventory. The thermal data were obtained at their original resolutions of 120 m (TM), 60

m (ETM+) and 100 m (TIRS) and then resampled to 30 m, and the seasonal composites included a different number of scenes and months in different years, which means that comparisons of absolute LST values between years are limited; in order to reduce this effect, normalised LST and within-year hotspots were used. The decision to use a 240 m grid and a 480 m neighbourhood also affects the size and shape of the hotspots, this being an example of the modifiable areal unit problem (Openshaw, 1984). Lastly, the action framework presented in Section 6 is based on the spatial evidence and on international practice; its measures and indicative targets have not yet been tested with local stakeholders. Future work should involve calibrating the coefficients using local energy and biomass data, test the sensitivity of the zones to grid size, incorporate higher-resolution imagery and in-situ atmospheric measurements, model future land-use scenarios, and co-develop the action portfolio with the city corporation, the development authority and the residents.

6. A Carbon-Conscious Heat Action Framework for Rajshahi

The findings indicate that in Rajshahi heat and carbon pressures are now spatially linked, though they are not the same everywhere. It would therefore be wrong to adopt a single heat plan for the entire city since different areas of the city require different combinations of measures. This section presents the evidence in the form of a structured framework which can be used by the city corporation and the development authority. The framework is set out as a four-step process: monitor, zone, act and review—with each step being supported by the satellite workflow described in this chapter (Figure 1-11).

During the monitoring phase, the values for LSCE, LST, NDVI and NDBI are updated annually on the basis of Landsat data that is freely available, using the Google Earth Engine and GIS scripts explained in Section 3, with a preference for using the pre-monsoon scenes so that the period of highest heat risk is included. In the zoning phase, the Gi* analysis reclassifies each 240 m cell into one of the five carbon-heat stress zones, thereby making the changes observable at the neighbourhood and street level. In the action phase, each zone is given a portfolio of measures taken from three areas: policy and governance, assessment and awareness, and tactical solutions. In the review phase, a small number of indicators are compared with the 2025 baseline and the portfolio is revised every five years, this being in line with the five-year interval of the benchmark years employed here.

6.1 Zone-Specific Action Portfolios

Table 1-7 summarises the recommended portfolio for each zone, based on the 2025 stress-zone map. Dual-stress zones receive the highest priority because every intervention there yields both a carbon and a cooling benefit. Carbon-only hotspots, which cover the largest stressed area, require measures that stop further loss of sink capacity even where heat is not yet statistically significant. Heat-only hotspots, largely on the exposed riverbank and the western fringe, call for shade and protection of people rather than carbon measures. Cool sinks,

by contrast, need protection above all, since they are the last places in the city where low emission and low temperature coincide.

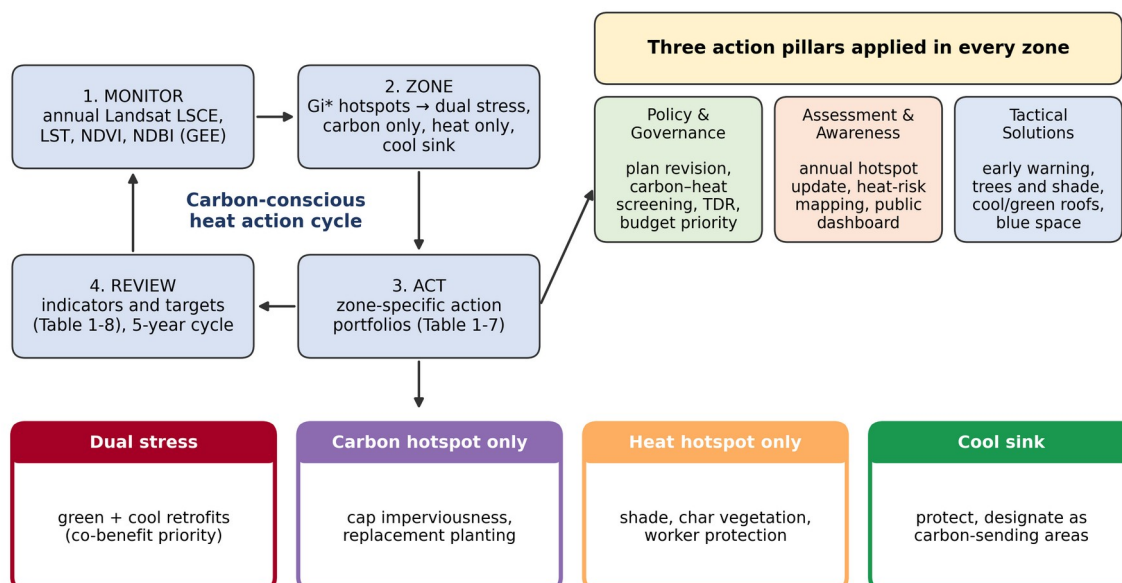


Figure 1-11. Carbon-conscious heat action framework for Rajshahi City Corporation, linking the monitor–zone–act–review cycle with three action pillars and zone-specific priorities.

Table 1-7. Zone-specific heat action portfolio for RCC based on the 2025 carbon–heat stress zones.

Zone (area 2025)	Main locations	Priority actions	Lead actors	Horizon
Dual stress (2.18 km ²)	North-central core; southern riverside	Street-tree and shade corridors; pocket parks on vacant plots; cool or green roofs on public buildings; restoration of ponds; shaded riverfront walkways	RCC, RDA, public works, ward committees	Short term (1–3 years)
Carbon hotspot only (9.59 km ²)	Central and west-central belt	Cap on impervious coverage per plot; mandatory replacement planting in building permits; compact, mixed-use redevelopment instead of peripheral sprawl	RDA (development control), RCC	Medium term (3–7 years)
Heat hotspot only (1.95 km ²)	Exposed char and riverbank; western fringe	Vegetative stabilisation of char land; shade structures at markets, bus stands and ghats; heat-safety measures for outdoor workers	RCC, Bangladesh Water Development Board, Forest Department	Short to medium term
Cool sink (3.53 km ²)	Eastern and north-eastern fringe	Protection from conversion; designation as carbon-sending areas with transfer of development rights; urban forestry and orchard conservation	RDA, RCC, Department of Environment	Immediate and continuous
Not significant (29.24 km ²)	Remaining city area	Preventive canopy targets; cool materials in new construction; annual watch for transitions into stress zones	RCC, RDA	Continuous

Note: RCC = Rajshahi City Corporation; RDA = Rajshahi Development Authority. Areas refer to the 861 analytical cells.

6.2 Policy and Governance

The easiest way in is through the current revision of the Rajshahi Metropolitan Development Plan, the earlier version of which spanned the period 2004 to 2024 (Rahman et al., 2024). The stress-zone maps could be included in the revised land-use zoning by designating the cool sinks as ecological or low-

intensity zones and marking the areas under dual stress as priority areas for retrofitting. In the context of development control, a simple step involving carbon and heat considerations could be introduced into the building-permit process: the applicant would have to report the change in LSCE resulting from the proposed land conversion and, in the case of stress zones, make up for the loss of sink capacity by carrying out on-site planting or by making a contribution to the city's greening fund. The send–receive framework from the companion study (Rahman et al., under review) offers the basis for a transfer-of-development-rights mechanism by which development potential is transferred from the eastern cool sinks to the carbon hotspots which are already developed.

The same logic can be applied to investment decisions. Since each unit of greening or cooling in a dual-stress cell tackles two problems simultaneously, such cells should be given first priority in the municipal budget and when applying for climate finance. The workflow is also cheap since it makes use of open data and free cloud computing, and a small geospatial and climate team within the city corporation, working in partnership with local universities such as RUET, would be enough to keep it going. Just as important as any individual physical measure is building this in-house capacity, since it enables the city to revise its priorities without having to depend on external consultants.

6.3 Assessment and Awareness

The current framework takes into account hazard (heat) and carbon pressure but has not yet considered human vulnerability. A reasonable next move for the city should be to superimpose the annual stress areas onto data regarding population density, informal settlements, schools, hospitals and places where outdoor work is carried out, thus creating a heat risk map at the ward level. Experience gained in Ahmedabad, India—where South Asia's first heat-health action plan integrated local evidence, vulnerability surveys and public outreach—demonstrates that such assessments are most effective when they are accompanied by communication to the affected groups (Knowlton et al., 2014). In Rajshahi, the maps could be made available on a basic public dashboard prior to each pre-monsoon season, incorporated into school and college awareness campaigns, and given to ward councillors so that residents can see how their own neighbourhood is changing. Community tree-adopting schemes in areas under dual stress would connect this increased awareness with actual action.

6.4 Tactical Solutions

Tactical measures can be divided into three categories. The first of these is early warning and response. The heat forecasts produced by the Bangladesh Meteorological Department can be turned into alerts at the level of individual wards, giving priority to wards under dual stress and those affected only by heat for access to drinking-water points, cooling shelters and for adjusting the working hours of outdoor labourers; the Ahmedabad plan demonstrated that a warning system triggered by a forecast can be put into place quickly and at a low cost (Knowlton et al., 2014). The second category involves physical cooling. It has been found that reflective and green roofs can reduce urban air temperatures, with an estimated decrease of about 0.3 K in the average ambient

temperature for each 0.1 increase in the city's albedo and as much as 3 K in the case of green roofs applied on a city scale (Santamouris, 2014), while urban trees lower the land surface temperature considerably when compared with built surfaces (Schwaab et al., 2021). Public buildings, schools and markets in areas under dual stress are appropriate first choices. The third group consists of blue and riverbank infrastructure. Restoring ponds that have been filled or degraded would provide both a carbon sink and a local cooling effect, and stabilising the exposed char land using grasses and trees would reduce the heat emitted from the riverbank without compromising its carbon sink function.

6.5 Monitoring Indicators and Review

Table 1-8 proposes a compact set of indicators that can be recomputed each year from the same satellite workflow. The 2025 values provide the baseline; the targets are indicative and should be confirmed with local stakeholders during the plan revision. Together, the indicators track whether the city is regaining sink capacity, whether stressed areas are shrinking and whether carbon and heat remain spatially coupled.

Table 1-8. Indicators for monitoring carbon-conscious heat action in RCC.

Indicator	Baseline (2025)	Indicative direction or target (2035)	Data source
Net land-specific carbon emission of RCC	$+3.71 \times 10^3 \text{ t CO}_2 \text{ yr}^{-1}$	Return to ≤ 0 (net sink)	LULC classification and LSCE
Vegetation cover	13.78 km ² (29.4%)	$\geq 35\%$ of city area	LULC classification
Share of net-source grid cells	67.0%	Below 50%	Grid LSCE
Dual-stress zone area	2.18 km ²	Reduce by half	Gi* stress zones
Cool-sink zone area	3.53 km ²	No net loss	Gi* stress zones
LST gap between source and sink cells	1.33 °C	Declining trend	Grid LST and LSCE
Heat-hotspot area within carbon hotspots	53%	Declining trend	Gi* stress zones

Note: targets are indicative proposals derived from the 2000–2025 trajectory (for example, vegetation cover reached 37.3% in 2020) and are intended as a starting point for local deliberation.

7. Conclusions

A satellite-based framework was developed in this chapter to connect land-specific carbon emissions with pre-monsoon land surface temperature and was then applied to Rajshahi City Corporation from 2000 to 2025. Between 2000 and 2025 the area covered by built-up land increased by 77.9% and vegetation decreased by 41.2%, causing the city to change from a net carbon sink to a net carbon source between 2010 and 2015 and leading to a net emission level of $+3.71 \times 10^3 \text{ t CO}_2 \text{ yr}^{-1}$ by 2025. On a grid scale, two-thirds of the city now acts as a net carbon source. There has been an increasing degree of coupling between carbon emissions and surface temperature: source areas are up to 1.4 °C warmer than sink areas, and since 2015 more than half of the city's significant heat has been concentrated in carbon hotspots, resulting in dual-stress zones in the north-central core and along the riverside. The coupling is nevertheless partial since exposed char land can at the same time be both a carbon sink and a heat source. By determining where carbon stress and heat exposure overlap, where they differ and where cool sinks still exist, the framework provides a transferable and low-cost approach to carbon-conscious urban heat management in Rajshahi and in other rapidly urbanising secondary

cities. Translating this evidence into practice, the chapter proposes a monitor–zone–act–review cycle in which every part of the city receives a portfolio of policy, awareness and tactical measures suited to its carbon and heat profile, together with indicators that can be updated each year from open satellite data. Protecting the eastern cool sinks, greening and cooling the dual-stress core, and embedding carbon and heat screening in the revision of the metropolitan development plan are the most immediate steps for Rajshahi.

Acknowledgements

This chapter is part of a research project funded by the Directorate of Research and Extension (DRE), Rajshahi University of Engineering & Technology (RUET), Bangladesh (Research Grant No. DRE/08/RUET/764(52)/Pro/2025–26/10(6)). The authors gratefully acknowledge the Directorate for its financial support and the Department of Urban & Regional Planning, RUET, for providing the academic environment, laboratory facilities and logistical support needed to carry out this study. The authors also thank the United States Geological Survey for providing Landsat data free of charge and Google for access to the Google Earth Engine platform.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this chapter.

References

- Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *J. R. Stat. Soc. Ser. B* 57, 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Cui, Y., Li, L., Chen, L., Zhang, Y., Cheng, L., Zhou, X., Yang, X., 2018. Land-use carbon emissions estimation for the Yangtze River Delta urban agglomeration using 1994–2016 Landsat image data. *Remote Sens.* 10, 1334. <https://doi.org/10.3390/rs10091334>
- Dewa, D.D., Buchori, I., 2023. Impacts of rapid urbanization on spatial dynamics of land use-based carbon emission and surface temperature changes in the Semarang Metropolitan Region, Indonesia. *Environ. Monit. Assess.* 195, 259. <https://doi.org/10.1007/s10661-022-10839-6>
- Dewan, A., Kiselev, G., Botje, D., Mahmud, G.I., Bhuian, M.H., Hassan, Q.K., 2021. Surface urban heat island intensity in five major cities of Bangladesh: Patterns, drivers and trends. *Sustain. Cities Soc.* 71, 102926. <https://doi.org/10.1016/j.scs.2021.102926>
- Fan, M., Wang, Z., Xue, Z., 2024. Spatiotemporal evolution characteristics, influencing factors of land use carbon emissions, and low-carbon development in Hubei Province, China. *Ecol. Inform.* 81, 102567. <https://doi.org/10.1016/j.ecoinf.2024.102567>

- Fang, J., Guo, Z., Piao, S., Chen, A., 2007. Terrestrial vegetation carbon sinks in China, 1981–2000. *Sci. China Ser. D Earth Sci.* 50, 1341–1350. <https://doi.org/10.1007/s11430-007-0049-1>
- Fattah, M.A., Morshed, S.R., Morshed, S.Y., 2021a. Impacts of land use-based carbon emission pattern on surface temperature dynamics: Experience from the urban and suburban areas of Khulna, Bangladesh. *Remote Sens. Appl. Soc. Environ.* 22, 100508. <https://doi.org/10.1016/j.rsase.2021.100508>
- Fattah, M.A., Morshed, S.R., Morshed, S.Y., 2021b. Multi-layer perceptron–Markov chain–based artificial neural network for modelling future land-specific carbon emission pattern and its influences on surface temperature. *SN Appl. Sci.* 3, 359. <https://doi.org/10.1007/s42452-021-04351-8>
- Getis, A., Ord, J.K., 1992. The analysis of spatial association by use of distance statistics. *Geogr. Anal.* 24, 189–206. <https://doi.org/10.1111/j.1538-4632.1992.tb00261.x>
- Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sens. Environ.* 202, 18–27. <https://doi.org/10.1016/j.rse.2017.06.031>
- Guerri, G., Crisci, A., Messeri, A., Congedo, L., Munafò, M., Morabito, M., 2021. Thermal summer diurnal hot-spot analysis: The role of local urban features layers. *Remote Sens.* 13, 538. <https://doi.org/10.3390/rs13030538>
- Gui, D., He, H., Liu, C., Han, S., 2023. Spatio-temporal dynamic evolution of carbon emissions from land use change in Guangdong Province, China, 2000–2020. *Ecol. Indic.* 156, 111131. <https://doi.org/10.1016/j.ecolind.2023.111131>
- Imran, H.M., Hossain, A., Shammas, M.I., Das, M.K., Islam, M.R., Rahman, K., Almazroui, M., 2022. Land surface temperature and human thermal comfort responses to land use dynamics in Chittagong city of Bangladesh. *Geomat. Nat. Hazards Risk* 13, 2283–2312. <https://doi.org/10.1080/19475705.2022.2114384>
- Kafy, A.-A., Al Rakib, A., Akter, K.S., Rahaman, Z.A., Faisal, A.-A., Mallik, S., Nasher, N.M.R., Hossain, M.I., Ali, M.Y., 2021. Monitoring the effects of vegetation cover losses on land surface temperature dynamics using geospatial approach in Rajshahi City, Bangladesh. *Environ. Chall.* 4, 100187. <https://doi.org/10.1016/j.envc.2021.100187>
- Knowlton, K., Kulkarni, S.P., Azhar, G.S., Mavalankar, D., Jaiswal, A., Connolly, M., Nori-Sarma, A., Rajiva, A., Dutta, P., Deol, B., Sanchez, L., Khosla, R., Webster, P.J., Toma, V.E., Sheffield, P., Hess, J.J., 2014. Development and implementation of South Asia's first heat-health action plan in

- Ahmedabad (Gujarat, India). *Int. J. Environ. Res. Public Health* 11, 3473–3492. <https://doi.org/10.3390/ijerph110403473>
- Li, L., Zhang, X., Yu, L., Xiao, H., 2024. Spatial-temporal evolution of land use carbon emissions and influencing factors in Zibo, China. *Front. Environ. Sci.* 12, 1472558. <https://doi.org/10.3389/fenvs.2024.1472558>
- Li, X., Liu, Z., Li, S., Li, Y., Wang, W., 2023. Urban land carbon emission and carbon emission intensity prediction based on patch-generating land use simulation model and grid with multiple scenarios in Tianjin. *Land* 12, 2160. <https://doi.org/10.3390/land12122160>
- Liu, Z., Han, L., Liu, M., 2024. High-resolution carbon emission mapping and spatial-temporal analysis based on multi-source geographic data: A case study in Xi'an City, China. *Environ. Pollut.* 361, 124879. <https://doi.org/10.1016/j.envpol.2024.124879>
- Lwasa, S., Seto, K.C., Bai, X., Blanco, H., Gurney, K.R., Kilkış, Ş., Lucon, O., Murakami, J., Pan, J., Sharifi, A., Yamagata, Y., 2022. Urban systems and other settlements, in: *Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, pp. 861–952. <https://doi.org/10.1017/9781009157926.010>
- Malakar, N.K., Hulley, G.C., Hook, S.J., Laraby, K., Cook, M., Schott, J.R., 2018. An operational land surface temperature product for Landsat thermal data: Methodology and validation. *IEEE Trans. Geosci. Remote Sens.* 56, 5717–5735. <https://doi.org/10.1109/TGRS.2018.2824828>
- Openshaw, S., 1984. *The Modifiable Areal Unit Problem. Concepts and Techniques in Modern Geography* 38. Geo Books, Norwich.
- Ord, J.K., Getis, A., 1995. Local spatial autocorrelation statistics: Distributional issues and an application. *Geogr. Anal.* 27, 286–306. <https://doi.org/10.1111/j.1538-4632.1995.tb00912.x>
- Qiu, S., Zhu, Z., Olofsson, P., Woodcock, C.E., Jin, S., 2023. Evaluation of Landsat image compositing algorithms. *Remote Sens. Environ.* 285, 113375. <https://doi.org/10.1016/j.rse.2022.113375>
- Rahman, M.N., Akter, K.S., Faridatul, M.I., 2024. Assessing the impact of urban expansion on carbon emission. *Environ. Sustain. Indic.* 23, 100416. <https://doi.org/10.1016/j.indic.2024.100416>
- Rahman, M.N., Faridatul, M.I., Masbah, M.H., Faruq, M.O., under review. Delineating carbon sending and receiving areas from land-specific carbon emission. Paper submitted to ICCIT 2026.
- Rahman, M.R., Mark, B.G., 2025. Geospatial analysis of urban warming: A remote sensing and GIS-based investigation of winter land surface

- temperature and biophysical composition in Rajshahi City, Bangladesh. *Sustainability* 17, 5107. <https://doi.org/10.3390/su17115107>
- Santamouris, M., 2014. Cooling the cities – A review of reflective and green roof mitigation technologies to fight heat island and improve comfort in urban environments. *Sol. Energy* 103, 682–703. <https://doi.org/10.1016/j.solener.2012.07.003>
- Schwaab, J., Meier, R., Mussetti, G., Seneviratne, S., Bürgi, C., Davin, E.L., 2021. The role of urban trees in reducing land surface temperatures in European cities. *Nat. Commun.* 12, 6763. <https://doi.org/10.1038/s41467-021-26768-w>
- Sresto, M.A., Siddika, S., Fattah, M.A., Morshed, S.R., Morshed, M.M., 2022. A GIS and remote sensing approach for measuring summer–winter variation of land use and land cover indices and surface temperature in Dhaka district, Bangladesh. *Heliyon* 8, e10309. <https://doi.org/10.1016/j.heliyon.2022.e10309>
- Srivastava, C., Bharat, A., 2025. Mapping carbon–thermal environments for comprehending real-time scenarios. *Acta Geophys.* 73, 933–953. <https://doi.org/10.1007/s11600-024-01387-3>
- Tiwari, A.K., Kanchan, R., 2024. Analytical study on the relationship among land surface temperature, land use/land cover and spectral indices using geospatial techniques. *Discov. Environ.* 2, 1. <https://doi.org/10.1007/s44274-023-00021-1>
- Tucker, C.J., 1979. Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sens. Environ.* 8, 127–150. [https://doi.org/10.1016/0034-4257\(79\)90013-0](https://doi.org/10.1016/0034-4257(79)90013-0)
- Ye, C., Ming, T., 2023. Land use carbon emissions estimation and carbon emissions control strategy effect scenario simulation in Zhejiang province. *Heliyon* 9, e20783. <https://doi.org/10.1016/j.heliyon.2023.e20783>
- Zha, Y., Gao, J., Ni, S., 2003. Use of normalized difference built-up index in automatically mapping urban areas from TM imagery. *Int. J. Remote Sens.* 24, 583–594. <https://doi.org/10.1080/01431160304987>
- Zhou, Y., Chen, M., Tang, Z., Mei, Z., 2021. Urbanization, land use change, and carbon emissions: Quantitative assessments for city-level carbon emissions in Beijing–Tianjin–Hebei region. *Sustain. Cities Soc.* 66, 102701. <https://doi.org/10.1016/j.scs.2020.102701>